MIL-H-51171F(EA)
2 February 1987
SUPERSEDING
MIL-H-51171E(EA)
27 June 1972

MILITARY SPECIFICATION

HEATER, WATER, LIQUID FUEL, SKID-MOUNTED, 600 GPH, M2

This specification is approved for use within Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of water heater.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4410

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-59-200 - Heater, Water, Liquid Fuel, Skid-Mounted, 600 GPH, M2.
5-45-3269 - Marking Diagram for Shipping Containers.

SPECIAL PACKAGING INSTRUCTION (SPI)

P5-59-200 - Heater, Water, Liquid Fuel, Skid-Mounted, 600 GPH, M2.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 396 - Standard Specification for Fuel Oils.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-59-200 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components All components shall conform to the specifications and drawings listed on Drawing 5-59-200 and subsidiary drawings.

3.2 Assembly. The heater shall be assembled in accordance with Drawing 5-59-200.

3.3 Leakage. The heater water system shall not leak when subjected to a water pressure of 225 ± 5 pounds per square inch gage (psig) for a minimum period of 5 minutes when tested as specified in 4.4.4.1.

3.4 Fuel. The heater shall be capable of using the following fuels:

(a) Fuel oil conforming to ASTM D 396, grade 1 or 2.
(b) Gasoline conforming to MIL-G-3056, type 1.

3.5 Performance. At an ambient temperature range of 0° to 125°F, the heater when tested as specified in 4.4.4.2 shall perform as follows:

(a) Using inlet water of 35° to 85°F the heater shall raise the temperature of the water at least 100°F within 5 minutes when the water is flowing at a rate of 600 ± 10 gallons per hour (gph).

(b) Thermostatic control shall stop burner if temperature of the water is between 188° and 212°F.

(c) The cutoff system shall operate and shut off fuel within 5 minutes when fuel does not ignite on demand.

(d) The outlet water temperature shall be capable of regulation within the following three operating ranges of flow and temperature and shall be automatically maintained within ±5°F of the regulated temperatures:

(1) 600 gph flow - temperature rise of 25° to 100°F above inlet temperature.
(2) 300 gph flow - temperature rise of 50° to 100°F above inlet temperature.
(3) 150 gph flow - temperature rise of 100°F only above inlet temperature.

3.6 Continuity. The main power cable assembly (Drawing 5-59-360) shall be continuous when tested in accordance with 4.4.4.3.

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3.7 Preproduction. Prior to the start of regular production, a preproduction sample of heaters shall be produced in accordance with this specification for examination and test (see 4.3).

3.8 Workmanship. The heater shall be free from contamination (foreign matter, oil, and grease) and damage such as cracks, dents, and deformation which could affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of one heater shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

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4.3.2 Inspection procedure.

4.3.2.1 For examination. The sample heater shall be examined in accordance with the classification of defects (4.4.3.3).

4.3.2.2 For tests. The sample heater shall be tested as specified in 4.4.4 using each of the fuels specified in 3.4.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the heaters of one interfix, number, produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4.2 Sampling for examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure.

4.4.3.1 For examination. The sample heaters and the level A unit pack requirements shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For tests. Each heater shall be tested in accordance with 4.4.4 using one of the specified fuels indicated in 3.4. The acceptance number shall be zero. Tests shall be conducted so that each of the two fuels shall be used sequentially; that is, whenever two successive heaters have been tested, both fuels will have been used.

4.4.3.3 Classification of defects.

(a) Heater, water, liquid fuel, skid-mounted, 600 gph, M2 (Drawing 5-59-200).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing
102	Component incorrectly assembled
103	Tooth type lock washers missing
104	Workmanship (3.8)

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Minor: AQL 2.5 percent defective

201	Marking illegible
202	Protective finish missing

(b) Cable assembly, main power (Drawing 5-59-360).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Continuity	4.4.4.3
102	Keyway position incorrect	

(c) Packaging inspection (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Packaging incorrect
102	Loose pack
103	Component missing
104	Container closure incorrect
105	Container damaged (broken or punctured)
106	Strapping loose or missing
107	Shroud missing or incorrect
108	Marking missing or incorrect

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Leakage. Prior to installing the safety valve to the boiler, use a 1/2 NPT pipe plug or other suitable closure device to configure the boiler for the water pressure test; and close the water inlet. After acceptable leakage test, remove closure device and install safety valve as required. Connect a suitable water supply to the water outlet. Apply the specified water pressure for the required period of time, and observe for water leakage at all joints and seams. This test must be performed on the boiler assembly prior to installation in the frame assembly and prior to assembly of the fire box to the boiler.

4.4.4.2 Performance.

(a) Install a suitable test thermometer, accurate to within ±1°F, in the water outlet line. Connect the water inlet of the heater to a suitable water supply of the specified temperature range. Connect the fuel inlet to a

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suitable fuel supply, and start the heater. With the water flowing at the specified rate, observe for the required water temperature rise on the test thermometer within the specified period of time.

(b) To check the thermostatic control cutoff, with water flow at 150 to 250 gph increase temperature control setting as required until the limit switch deactivates burner. Record the outlet water temperature when the limit switch deactivates burner.

(c) With the heater operating, disconnect the fuel suction line from the fuel supply. The cutoff system should operate and shut off the heater within the specified time.

(d) With the test thermometer in the water outlet line, regulate the temperature selector to obtain successively three separate temperatures, each to be within one of the three operating ranges of flow and temperature. After temperature is stabilized, observe the test thermometer to see that the heater will maintain the temperature of the outlet water within the specified tolerance for 5 minutes without adjustment. Each heater shall be tested at different temperatures within each of the three operating ranges of flow and temperature.

4.4.4.3 Continuity. Check the cable assembly with an ohmmeter.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in SPI P5-59-200.

6. NOTES

6.1 Intended use. This item in intended for use with a decontaminating apparatus and for other purposes requiring a heater for a supply of hot water.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of unit packing and packing required.
(c) Color of finish of water heater required by using service.
(d) Additional identification plates if required by using service.
(e) Preproduction.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(f) Mark in accordance with M12A1 Decontaminating Apparatus, MIL-D-51091, when procured as a component thereto.

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6.3 Subject term (key word) listing.

Boiler
Decontaminating apparatus
Decontaminating equipment
Heater, water
Heater, water, liquid fuel
Heater, water, liquid fuel, skid-mounted
M2

6.4 Changes from previous issues. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - EA

Preparing activity:

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Project 4410-A062

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-51171F(EA)	HEATER, WATER, LIQUID FUEL, SKID-MOUNTED, 600 GPH, M2

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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